

Set Theory SoSe 25-26, Wise 25-26

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2 On Hilbert's First Question

Definition 2.1 (Continuum Hypothesis). The Continuum Hypothesis(CH) asserts that there does not exist an uncountable subset A of $\mathbb{R}$ such that $\overline{\overline{N}} < \overline{\overline{A}} < \overline{\overline{\mathbb{R}}}$, i.e., $\aleph_1 = 2^{\aleph_0}$.

Definition 2.2 (Generalized Continuum Hypothesis). The Generalized Continuum Hypothesis(GCH) asserts that for all $\alpha < \text{Ord}$, $2^{\aleph_\alpha} = \aleph_{\alpha+1}$.

Kurt Gödel in 1938 showed that $\text{Con}(\text{ZFC}) \rightarrow \text{Con}(\text{ZFC} + \text{GCH})$ using his constructible universe(L), but in this section we will give a ZFC model where CH holds by forcing.

2.1 A ZFC model in which CH fails

Definition 2.3. Let $I \neq \emptyset$, define $\mathbb{C}(I) = \{p : p \text{ is a function } \text{dom}(p) \rightarrow \omega, \text{dom}(p) \subseteq I \times \omega, \overline{\overline{\text{dom}(p)}} < \aleph_0\}$, ordered by reverse inclusion, i.e., product of $\overline{\overline{I}}$ -many Cohen forcing with finite support.

Observations: $\mathbb{C}(I)$ has the c.c.c. and $1 \Vdash \overline{\overline{I}} \leq 2^{\aleph_0}$.

Theorem 2.4 (Paul Cohen, 1963). *If ZFC is consistent then so is $\text{ZFC} + \neg\text{CH}$. In fact $\text{Con}(\text{ZFC}) \rightarrow \text{Con}(\text{ZFC} + 2^{\aleph_0} = \aleph_2)$*

Proof. From now on always assume M is a countable transitive model of ZFC. We may assume $M \models \text{CH}$. We force with $\mathbb{C}(\omega_2^M)$. Let $G \subseteq \mathbb{C}(\omega_2^M)$ be a generic filter over M . As $\mathbb{C}(\omega_2^M)$ has the c.c.c. so $\omega_2^M = \omega_2^{M[G]}$.

Since $1 \Vdash \overline{\overline{I}} < 2^{\aleph_0}$, there exists a injection $F : \omega_2 \rightarrow \omega^\omega \cap M[G]$ in $M[G]$ so $M[G] \models 2^{\aleph_0} \geq \aleph_2$ thus $M[G] \models \neg\text{CH}$. Now we show $M[G] \models 2^{\aleph_0} \leq \aleph_2$. Let $x \in \omega^\omega \cap M[G]$ and say $x = \tau^G$, define $E_n = \{p \in \mathbb{C}(\omega_2^M) : \exists m < \omega, p \Vdash \tau(\check{n}) = \check{m}\}$ for $n < \omega$. Since E_n is dense, let $A_n \subseteq E_n$ be an maximal antichain and define $\sigma = \{([\check{n}, \check{m}], p) : p \in A_n, p \Vdash \tau(\check{n}) = \check{m}\}$ where $[\rho, \sigma] = \{(\{(\rho, 1), (\sigma, 1)\}, 1), (\{(\rho, 1)\}, 1)\}$ for $y = \tau^G, z = \sigma^G$, in fact $[\rho, \sigma]^G = (y, z)$. We claim $x = \tau^G = \sigma^G$. If $(n, m) \in \sigma^G$ then $\exists p \in G$ such that $([\check{n}, \check{m}], p) \in \sigma, p \Vdash \tau(\check{n}) = \check{m}$, so $(n, m) \in \tau^G$. For the other side $\tau^G \subseteq \sigma^G$, the proof is similar. So every real $x \in \omega^\omega \cap M[G]$ has a "nice" name. Notice that for any $n < \omega$, there are countably many m and p such that $([\check{n}, \check{m}], p) \in \sigma, \sigma \subseteq \omega \times \omega \times \mathbb{C}(\omega_2^M)$. Thus there are only $(\omega_2^M)^{\aleph_0}$ -many nice names in M , by Hausdorff's formula we obtain: $(\omega_2^{\aleph_0})^M = (2^{\aleph_0} \cdot \omega_2)^M \stackrel{\text{CH}}{=} (\omega_1 \cdot \omega_2)^M = (\omega_2)^M$. Since there are $\omega_2^M = \omega_2^{M[G]}$ many names, $M[G] \models 2^{\aleph_0} = \aleph_2$. ■

However, we are not done yet. We want to get rid of the transitivity hypothesis.

Theorem 2.5 (Levy-Montague Reflection Principle). *Let $\varphi(x, \vec{y}) \in \mathcal{L}_\in$ and $\vec{b}$ be sets, then there is a club class C of ordinals such that for any $\alpha \in C, \vec{b} \in V_\alpha$ and for all $a \in V_\alpha$ we have $V_\alpha \models \varphi(a, \vec{b}) \iff V \models \varphi(a, \vec{b})$*

2.2 A ZFC model in which CH holds

Idea: Add a Cohen subset of ω_1 . Let $\mathbb{C}_{\omega_1} := \mathbb{C}_{\omega_1}(1) := \text{Add}(\omega_1, 1) = {}^{<\omega_1}\omega_1$ and order it by inverse of end-extension (we use the Cohen forcing at ω_1 .)

Notice that $\overline{<\omega_1\omega_1} = \Sigma_{\alpha<\omega_1} \overline{\alpha\omega_1} = \Sigma_{\alpha<\omega_1} \omega_1^{\aleph_0} = \omega_1 \cdot \omega_1^{\aleph_0} = \omega_1 \cdot \omega_1 \cdot 2^{\aleph_0} = 2^{\aleph_0}$. Thus $\mathbb{C}_{\omega_1}$ has the $2^{\aleph_0+}$ -c.c.

Now we fix $G \subseteq \mathbb{C}_{\omega_1}$ a generic filter over V .

Definition 2.6. Let $\mathbb{P}$ be a poset and κ a regular cardinal, we say $\mathbb{P}$ is $<\kappa$ -closed iff $\forall \lambda < \kappa$ and $\forall (p_\alpha : \alpha < \lambda)$ such that $p_\alpha \in \mathbb{P}$ for all α and for all $\alpha < \beta < \lambda, p_\beta \leq p_\alpha$, there exists $q \in \mathbb{P}$ such that $q \leq p_\alpha$ for all $\alpha < \lambda$.

Lemma 2.7. Let $\mathbb{P}$ be a $<\kappa$ -closed poset and $g \subseteq \mathbb{P}$ be generic. Then ${}^{<\kappa}V \cap V = {}^{<\kappa}V \cap V[g]$ thus it also preserves cardinals $\leq \kappa$.

Proof. Let $\lambda < \kappa, f: \lambda \rightarrow V$ and say that $f = \tau^g \in V[g]$. We want to show $f \in V$. Let $X \in V$ be such that $\text{ran}(f) \subseteq X$ and let $p \in g$ such that $p \Vdash \tau: \check{X} \rightarrow \check{X}$ is a function". Say $q \in \mathbb{P}$ decides τ if and only if for all $\alpha < \lambda, \exists b \in X$ such that $q \Vdash \tau(\check{\alpha}) = \check{b}$.

We want to show that $D = \{q \in \mathbb{P} : q \text{ decides } \tau\}$ is dense below p : take any $r \leq p$ and we construct a sequence $(q_\alpha : \alpha \leq \lambda)$ as follows:

- set $q_0 = r$,
- suppose q_α has already been constructed, define $q_{\alpha+1} \leq q_\alpha$ by $\exists b \in X$ such that $q_{\alpha+1} \Vdash \tau(\check{\alpha}) = \check{b}$, by fullness, such a condition $q_{\alpha+1}$ existed,
- at limit step, by $<\kappa$ -closedness, we set $q_\alpha \leq q_\beta$ for limit α and $\forall \beta < \alpha$.

Now $q_\lambda \leq r$ decides τ . Thus $q \in D \cap G \neq \emptyset$, let $f_q: \lambda \rightarrow X$ be the function that $q \Vdash f_q(\check{\alpha}) = \tau(\check{\alpha})$ and thus $f = f_q \in V$. ■

Note that $\mathbb{C}_{\omega_1}$ is $<\aleph_1$ -closed as ω_1 is regular thus $\omega_1^V = \omega_1^{V[G]}$ and $\omega^\omega \cap V = \omega^\omega \cap V[g]$. In order to show $\text{Con}(\text{ZFC} + \text{CH})$, we need to show $V[G] \models 2^{\aleph_0} = \aleph_1$, i.e., $V[G] \models \exists$ a surjection $\omega_1 \rightarrow \omega^\omega$.

Proof. Let $F = \bigcup G: \omega_1 \rightarrow \omega_1$, now we construct a surjective function $\hat{F}: \omega_1 \rightarrow \omega^\omega$ in $V[G]$.

Define

$$\hat{F}(\alpha)(n) = \begin{cases} F(\omega \cdot \alpha + n) & \text{if } F(\omega \cdot \alpha + n) < \omega \\ 0 & \text{otherwise} \end{cases}$$

We claim that $\hat{F}$ is onto: let $x \in \omega^\omega$, define $D_x = \{p \in \mathbb{C}_{\omega_1} : \exists \alpha < \omega_1, \forall n < \omega, p(\omega \cdot \alpha + n) = x(n)\} \in V$, it is easy to see D_x is dense. Let $p \in D_x \cap G$, then there exists $\alpha < \omega_1$ such that for any $n < \omega, p(\omega \cdot \alpha + n) = x(n) = F(\omega \cdot \alpha + n) = F(\omega \cdot \alpha + n) = \hat{F}(\alpha)(n)$ thus $\hat{F}(\alpha) = x$, $\hat{F}$ is onto.

Therefore $V[G] \models \text{CH}, \text{Con}(\text{ZFC}) \rightarrow \text{Con}(\text{ZFC} + \text{CH})$. ■

We have settled the continuum problem.